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**COURSE : Fundamentals of Data Science**

**CODE : DSA0405**

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### **CO3-AT1 Case Study Assignment Solutions**

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#### **Case Study 1 — Employee Salaries:**

**Data (Annual Salaries in ₹ lakhs):**

4, 5, 5, 6, 7, 8, 8, 9, 30 ( $N = 9$ )

##### **1. Mean, Median, and Mode**

- **Mean ( $\mu$ ):**

$$\mu = \frac{\sum X}{N} = \frac{4 + 5 + 5 + 6 + 7 + 8 + 8 + 9 + 30}{9} = \frac{82}{9} \approx \mathbf{9.11 \text{ lakhs}}$$

- **Median:**

Since  $N = 9$  (odd), the median is the  $\frac{9+1}{2} = 5^{th}$  ordered value:

**Median = ₹7 lakhs**

- **Mode:**

The values 5 and 8 both appear twice.

**Mode = ₹5 lakhs and ₹8 lakhs (Bimodal)**

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##### **2. Range**

$$\text{Range} = \text{Maximum} - \text{Minimum} = 30 - 4 = \mathbf{26 \text{ lakhs}}$$

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##### **3. Variance and Standard Deviation (Population formulas)**

- **Deviations from Mean ( $\mu = 9.11$ ):**

- $(4 - 9.11)^2 = 26.11$

- $(5 - 9.11)^2 \times 2 = 16.89 \times 2 = 33.78$
- $(6 - 9.11)^2 = 9.67$
- $(7 - 9.11)^2 = 4.45$
- $(8 - 9.11)^2 \times 2 = 1.23 \times 2 = 2.46$
- $(9 - 9.11)^2 = 0.01$
- $(30 - 9.11)^2 = 436.39$

- **Sum of Squared Differences ( $\sum(X - \mu)^2$ ):**

$$26.11 + 33.78 + 9.67 + 4.45 + 2.46 + 0.01 + 436.39 = 512.87$$

- **Population Variance ( $\sigma^2$ ):**

$$\sigma^2 = \frac{512.87}{9} \approx \mathbf{56.99}$$

(Sample Variance  $s^2 = \frac{512.87}{8} \approx 64.11$ )

- **Standard Deviation ( $\sigma$ ):**

$$\sigma = \sqrt{56.99} \approx \mathbf{7.55 \text{ lakhs}}$$

(Sample  $s = \sqrt{64.11} \approx 8.01 \text{ lakhs}$ )

#### 4. Outlier Identification

- **Outlier Value: ₹30 lakhs**
- *Verification:*

$$Q_1 = 5, Q_3 = 8, \text{IQR} = 3.$$

$$\text{Upper Fence} = Q_3 + 1.5 \times \text{IQR} = 8 + 1.5(3) = \mathbf{12.5}.$$

Since  $30 > 12.5$ , **30 is an outlier.**

#### 5. Best Measure of Central Tendency

**Median (₹7 lakhs)** best represents the typical salary.

- **Explanation:** The mean (₹9.11 lakhs) is heavily distorted upwards by the extreme outlier of ₹30 lakhs, making it higher than 89% of all actual employee salaries. The median remains robust against outliers and accurately reflects the central core of the dataset.

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## Case Study 2 — Website Response Time:

### Data (Response Time in ms):

120, 125, 128, 130, 132, 135, 140, 145, 150, 300 ( $N = 10$ )

#### 1. Mean and Median

- **Mean ( $\mu$ ):**

$$\mu = \frac{120 + 125 + 128 + 130 + 132 + 135 + 140 + 145 + 150 + 300}{10} = \frac{1395}{10} \\ = \mathbf{139.5 \text{ ms}}$$

- **Median:**

Average of the 5<sup>th</sup> (132) and 6<sup>th</sup> (135) values:

$$\text{Median} = \frac{132 + 135}{2} = \mathbf{133.5 \text{ ms}}$$

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#### 2. Range

$$\text{Range} = 300 - 120 = \mathbf{180 \text{ ms}}$$

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#### 3. Variance and Standard Deviation

- **Sum of Squared Differences ( $\sum(X - \mu)^2$ ):**

$$(120 - 139.5)^2 + (125 - 139.5)^2 + \dots + (300 - 139.5)^2 = 27792.5$$

- **Population Variance ( $\sigma^2$ ):**

$$\sigma^2 = \frac{27792.5}{10} = \mathbf{2779.25}$$

(Sample  $s^2 = 3088.06$ )

- **Standard Deviation ( $\sigma$ ):**

$$\sigma = \sqrt{2779.25} \approx \mathbf{52.72 \text{ ms}}$$

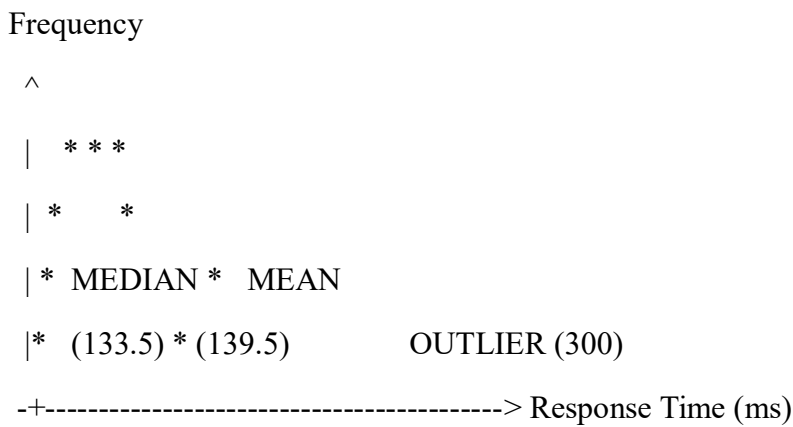
(Sample  $s \approx 55.57 \text{ ms}$ )

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#### 4. Skewness

The data is **Right-Skewed (Positively Skewed)** because the mean (139.5 ms) is greater than the median (133.5 ms), caused by the large extreme response time of 300 ms.

Data Distribution Curve (Right-Skewed):



### 5. Why Median is More Useful

The median is not sensitive to extreme, temporary performance spikes (like 300 ms). It provides a reliable estimate of what a standard user typically experiences, whereas the mean overestimates standard response times.

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### Case Study 3 — Student Performance:

Data (Test Marks out of 100):

42, 45, 48, 50, 52, 52, 55, 58, 60, 62, 65, 95 ( $N = 12$ )

#### 1. Quartiles ( $Q_1$ , $Q_2$ , $Q_3$ )

- $Q_2$  (Median): Average of 6<sup>th</sup> and 7<sup>th</sup> terms  $\rightarrow \frac{52+55}{2} = 53.5$
  - $Q_1$  (Lower Half: 42, 45, 48, 50, 52, 52): Average of 3<sup>rd</sup> and 4<sup>th</sup> terms  $\rightarrow \frac{48+50}{2} = 49$
  - $Q_3$  (Upper Half: 55, 58, 60, 62, 65, 95): Average of 3<sup>rd</sup> and 4<sup>th</sup> terms  $\rightarrow \frac{60+62}{2} = 61$
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#### 2. Interquartile Range (IQR)

$$IQR = Q_3 - Q_1 = 61 - 49 = 12$$

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### 3. Outlier Identification ( $1.5 \times \text{IQR}$ Rule)

- $1.5 \times \text{IQR} = 1.5 \times 12 = 18$
- **Lower Boundary:**  $Q_1 - 18 = 49 - 18 = 31$
- **Upper Boundary:**  $Q_3 + 18 = 61 + 18 = 79$

Outlier Check:  $95 > 79 \rightarrow 95$  is an Outlier.

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### 4. Mean and Median

- **Mean ( $\mu$ ):**

$$\mu = \frac{684}{12} = 57$$

- **Median:**

$$\text{Median} = 53.5$$

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### 5. Impact of Outlier on Mean

The high score of 95 pulls the mean mark up to 57, which is higher than the performance of over half the class. Without 95, the mean for the remaining 11 students would be approximately 48.55.

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## Case Study 4 — E-Commerce Orders:

### Data (Daily Orders over 10 days):

18, 20, 22, 22, 24, 25, 26, 28, 30, 35 ( $N = 10$ )

#### 1. Mean, Median, and Mode

- **Mean ( $\mu$ ):**

$$\mu = \frac{250}{10} = 25 \text{ orders}$$

- **Median:**

$$\text{Median} = \frac{24 + 25}{2} = 24.5 \text{ orders}$$

- **Mode:**

**Mode = 22 orders** (appears twice)

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## 2. Variance and Standard Deviation

- **Sum of Squared Differences ( $\sum(X - \mu)^2$ ):**

$$\begin{aligned}(18 - 25)^2 + (20 - 25)^2 + \dots + (35 - 25)^2 \\ = 49 + 25 + 9 + 9 + 1 + 0 + 1 + 9 + 25 + 100 = 228\end{aligned}$$

- **Population Variance ( $\sigma^2$ ):**  $\frac{228}{10} = \mathbf{22.8}$
- **Population Standard Deviation ( $\sigma$ ):**  $\sqrt{22.8} \approx \mathbf{4.77}$  orders

(Sample  $s^2 = 25.33$ ,  $s \approx 5.03$ )

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## 3. Coefficient of Variation (CV)

$$CV = \left(\frac{\sigma}{\mu}\right) \times 100 = \left(\frac{4.77}{25}\right) \times 100 \approx \mathbf{19.08\%}$$

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## 4. Meaning of Standard Deviation for the Company

A standard deviation of  $\approx 4.77$  orders indicates that daily order volumes typically deviate by about 5 orders from the average daily demand of 25. This helps inventory and logistics teams plan safety stock levels to absorb daily fluctuations.

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## 5. Variability Assessment

With a Coefficient of Variation of **19.08%** (which is well below 30%), the order volume can be described as **moderately low / stable in variability**.

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## Case Study 5 — Comparing Two Data Science Models:

- **Model A:** 2, 3, 3, 4, 4, 5, 5, 6 ( $N = 8$ )
- **Model B:** 1, 1, 2, 4, 5, 7, 8, 9 ( $N = 8$ )
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Summary Comparison Table:

| Metric                                | Model A | Model B | Formula / Method             |
|---------------------------------------|---------|---------|------------------------------|
| Sum of Errors ( $\sum X$ )            | 32      | 37      | Add all values               |
| 1. Mean ( $\mu$ )                     | 4.0     | 4.625   | $\frac{\sum X}{N}$           |
| Sum of Squared Diff.                  | 12      | 63.875  | $\sum (X - \mu)^2$           |
| 2. Population Variance ( $\sigma^2$ ) | 1.5     | 7.984   | $\frac{\sum (X - \mu)^2}{N}$ |
| 3. Standard Deviation ( $\sigma$ )    | 1.22    | 2.83    | $\sqrt{\sigma^2}$            |
|                                       |         |         |                              |

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4. Which Model Has More Consistent Predictions?

**Model A** is significantly more consistent. Its standard deviation ( $\sigma = 1.22$ ) is much lower than Model B's ( $\sigma = 2.83$ ), indicating that its error rates do not fluctuate wildly across predictions.

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5. Is Lower Mean Error Always Better?

**No, lower mean error alone does not make a model superior.**

- Explanation:** While Model A happens to have both lower mean error and higher consistency here, in general, a model with a slightly lower mean error could have very high variance (extreme occasional failures). In operational settings, high volatility

makes models unpredictable and risky. Model evaluation must always balance average performance (mean error) with stability (variance/standard deviation).